Replication of the orthogonal latent-mean contrasts in Jeffries (2024)

27 October 2026

Abstract

When four groups have unequal size, do Jeffries's three latent-mean contrasts stay orthogonal, and do the Monte Carlo averages sit on the scale of the data-generating means? Jeffries (2024) defines three codes for four latent means, states that imbalance makes the set non-orthogonal, and reports multigroup and MIMIC averages near -0.40, -0.20, and 0 when the generating means are $1, 0.6, 0.8, 0.4$. We proved the weighted products of the printed codes and the population values of the printed contrast formulas, and we refit the one-factor model in lavaan 0.6.17. The codes are orthogonal when the groups are equal. Under Jeffries's imbalance pattern only the product of the first and third codes is nonzero. On the generating means the contrasts are $-2/5$, $-1/5$, and 0. With the first loading fixed at 1, the same formulas give $-7/25$, $-7/50$, and 0, and the Monte Carlo averages follow that scale. Fixing the loadings at 0.7, or fixing the factor variance at 0.50, brings the averages back near the printed table. Percent significant is close to Tables 4.1 and 4.2 under either scale. The analysis script that would decide the identification is not in the dissertation.

1 What this replication did

This section answers what we contributed and what we took from Jeffries. We contributed the orthogonality proofs, the population arithmetic, a lavaan replay, and Table 1. Jeffries contributed the common-factor design, the contrast codes, the simulation conditions, and Tables 4.1 through 4.3. The crops in the later sections are photographs of that dissertation. Every other number is either a theorem in `LatentContrasts/Orthogonality.lean` or a summary of `simulation/out/`.

The dissertation studies one factor with four indicators. Four groups have latent means $\alpha = (1, 0.6, 0.8, 0.4)$ in the true-difference condition and $\alpha = (1, 1, 1, 1)$ in the null. Strong invariance is built into the population. Total sample size is 240, 480, or 960. The group shares are 25/25/25/25, 35/35/15/15, and 45/45/5/5. Jeffries asks whether contrasts that are orthogonal at equal sizes stay accurate when the sizes differ, and compares a multigroup confirmatory factor model with a MIMIC model.

2 Comparison

Table 1 puts one published claim on each row. "Match" means the printed value and our value agree as exact numbers or as the same Monte Carlo target. "Close" means our Monte Carlo, at 200 replicates, sits within ordinary sampling error of the printed average or percent. "Partial" means one piece of the claim holds and another does not. "No match" means the two numbers are on different scales after the checks in Sections 3 and 4.

Claim	Published	Ours	Verdict
Unweighted products of the Table 2.2 codes	Equation (2.4) is zero	Proved zero	match
Contrasts on $(1, 3/5, 4/5, 2/5)$	-0.4, -0.2, 0	$-2/5$, $-1/5$, 0	match
Third contrast while the means differ	Used as a null contrast	Proved equal to 0	match
Imbalance destroys orthogonality	The set becomes non-orthogonal	Only code 1 with code 3 fails	partial
Marker scale of those contrasts	Tables give about -0.40, -0.20, 0	$-7/25$, $-7/50$, 0, and the fits follow it	no match
Loadings fixed at 0.7, $N =$ 20,000 per group	About -0.40, -0.20, 0	-0.398, -0.194, 0.001	close
Factor variance fixed at 0.50, same sample	About -0.40, -0.20, 0	-0.400, -0.195, 0.001	close
Table 4.1, $N = 240$, equal groups, percent significant	87.23, 35.57, 5.29	Marker model, 200 replicates: 90.5, 37.5, 4.5	close
Table 4.1, $N = 240$, shares 45/45/5/5, percent significant	49.23, 21.08, 7.77	52.0, 22.5, 4.5	partial
Table 4.2 MIMIC averages, $N = 240$, equal groups	-0.400, -0.200, 0.001	Marker MIMIC: -0.282, -0.136, 0.002	no match
Same MIMIC cell, loadings fixed at 0.7	-0.400, -0.200, 0.001	-0.406, -0.203, -0.003	close
Numeric codes ± 1 as covariates, loadings fixed	The full contrast	About half: -0.195, -0.095, -0.007	no match
Table 4.3, $N = 240$, shares 45/45/5/5, percent significant	8.13, 8.59, 7.73	Marker null model: 8.5, 5.5, 9.0	partial

Table 1: Published claims against this replication. Monte Carlo rows use 200 replicates except the two large-sample rows, which are one sample of 20,000 per group.

The unweighted-product row is the identity in equation (2.4), proved at 103. The population row is equations (3.3)–(3.5) on the printed means, proved at 123. The third contrast is zero while the first two means differ, proved at 127. The imbalance row is partial because 68 shows that the pattern of two equal large groups and two equal small groups leaves the products of codes 1 with 2, and 2 with 3, at zero, and sets the remaining product to $2(b - a)$. At the $N = 240$ sizes that product is -96 in condition 2 and -192 in condition 3.

The marker row is a no match for the printed averages. Chapter III fixes the first loading at 1 and frees the factor variances. Under the data-generating loading 0.7, the contrasts on that scale are $-7/25$, $-7/50$, and 0, proved at 131. Two hundred marker replicates at $N = 240$ in the equal true-difference cell averaged -0.286, -0.134, and 0.007. The printed Table 4.1 cell averages -0.4046, -0.2027, and -0.0010. The page crop and the text layer agree on both the marker sentence and the table, so the gap is not an extraction error. Turning the marker constraint off, by fixing every loading at 0.7 or by fixing the factor variance at 0.50, moves our averages onto the printed scale. Those two rows are therefore close, and the marker row stays a no match. The same split appears in the MIMIC block: a marker MIMIC misses the printed averages, and a MIMIC with loadings fixed at 0.7 is close.

Percent significant can agree while the averages do not. Multiplying a contrast and its standard

error by 0.7 leaves the z statistic unchanged. The equal true-difference percents in Table 4.1 are close to the marker replay. In the 45/45/5/5 cell the first two percents are close and the third, 4.5 against 7.77, is not, at 200 replicates, so that row is partial. The null percents in Table 4.3 are partial for the same reason: 8.5 is close to 8.13, while 5.5 and 9.0 sit off 8.59 and 7.73. A binomial standard error at 200 replicates and a rate near 0.08 is about two percentage points.

The numeric-code row is a no match to the printed averages. With the codes stored as numbers, the slope is about half the defined contrast, because the code changes by 2. With the codes stored as factors whose levels are -1 and 1, lavaan's treatment contrast estimates the full difference. The appendix stores the codes as factors. The half-size slopes are what the numeric coding does, not what Table 4.2 prints.

3 Assumptions

Each subsection is a choice the dissertation does not pin down and that this replay supplied.

3.1 Random-number stream

Jeffries sets `set.seed(1313)` and passes seed 13131 to `exportData`. We drew normal factor scores and normal residuals in R with the L'Ecuyer generator and `set.seed(13131)`. The replicates are not Jeffries's replicates. Agreement is about the Monte Carlo mean, not about a shared sample.

3.2 R version and the optimizer

The dissertation names R 4.2.2, simsem 0.5.16, and lavaan 0.6.17, with estimator MLR and up to 10,000 iterations. We used R 4.3.3 and lavaan 0.6.17. We did not call simsem. The optimizer is lavaan's default `nlminb`, with `iter.max` set to 1,000. A fit counts as a success when it converges and the three contrast standard errors are finite.

3.3 Replicate count

Jeffries keeps 10,000 converged replicates in each cell, after drawing 15,000. We used 200 replicates for every $N = 240$ cell in the main grid and in the identification check, and 100 replicates at $N = 960$. The large-sample rows are one dataset with 20,000 observations in each group. Percents from 200 replicates are coarser than percents from 10,000.

3.4 Intercept of the first indicator

Appendix A writes `i1 ~ 1*` and `i2 ~ 1*1`, and the same `1*1` form for the third and fourth indicators. lavaan 0.6.17 rejects `i1 ~ 1*` as a formula. We fixed every intercept at 1, which is the value printed for the other three indicators and the value named in the population description.

3.5 Sign of the third code

Table 2.2 codes the third contrast as $+1, -1, -1, +1$. Appendix A codes it as $-1, +1, +1, -1$. We used the table. Section 4.2 records why the population value does not depend on that choice.

3.6 Data-generating variances

The population paragraph says the loading 0.70 accounts for 49% of the indicator variance. The appendix sets the loading to 0.7, the factor variance to 0.5, and the residual variance to 0.51. We generated data from those three appendix numbers. The squared loading 0.49 is then the explained variance only if the factor variance is 1 and the residual is $1 - 0.49$. That is not the appendix.

4 Ambiguities that were resolved

4.1 The ink matches the text layer

The question was whether a sign or a printed average in the text extract disagreed with the page. It does not, on the pages that carry the claims in Table 1.

The artifact is the set of PNG crops in `formalization/latent-contrasts/paper/figures/`. The script is `paper/figures/crop_pages.py`. It runs `pdftoppm -png -r 600`. Pages whose PDF rotation flag is 90 degrees are turned clockwise into reading order before the crop. The command is `python3 formalization/latent-contrasts/paper/figures/crop_pages.py`, and the output is those PNG files. The commit that contains the script is 2dcc1c414311c6a3a4d20cfe25ac390fb97a50cd.

Table 2.2
Planned Orthogonal Contrast Code Matrix

Group	C_1	C_2	C_3
1	-1	-1	1
2	1	-1	-1
3	-1	1	-1
4	1	1	1
Σ	0	0	0

Figure 1: Table 2.2. Codes $C_1 = (-1, 1, -1, 1)$, $C_2 = (-1, -1, 1, 1)$, $C_3 = (1, -1, -1, 1)$.

The standard-error column in these tables is about 0.001 to 0.002. That is the standard error of a mean of 10,000 replicates, not the average within-replicate standard error. Our column `sd_est` is the standard deviation across replicates, about 0.08 to 0.14 at $N = 240$, and is not compared with that printed column.

4.2 The third code in the appendix

The question was whether Appendix A uses the Table 2.2 sign for the third code. It uses the opposite sign.

The artifact is `figures/appendix_c3.png`, produced by the same script, output, and commit as the previous subsection. On the printed means the third contrast is zero, so changing the sign leaves the population value at zero and does not move a percent-significant column for a two-sided test of that contrast. The replay uses the table's sign.

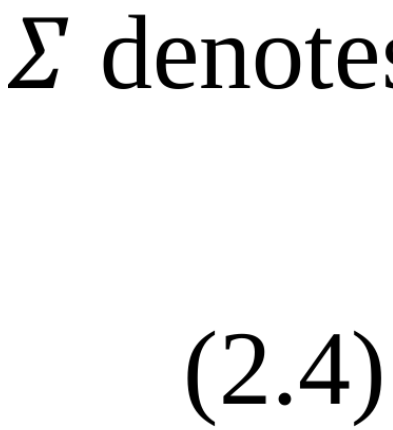


Figure 2: Equation (2.4). The three unweighted cross-products are zero.

4.3 Factor coding against numeric coding

The question was whether a covariate coded ± 1 estimates the contrast in equations (3.3)–(3.5) or half of it. As a factor with levels -1 and 1, lavaan's treatment contrast estimates the full difference. As a number, the slope is half of that difference, because the code changes by 2.

The check is `simulation/identification_check.R`, model `mimic_numeric`, output `simulation/out/identification_check.csv`. The command is `Rscript identification_check.R 200 out` from `formalization/latent-contrasts/simulation`. The commit is 2dcc1c414311c6a3a4d20cfe25ac390fb97a50cd. At $N = 240$, equal groups, loadings fixed at 0.7, and 200 replicates, the numeric slopes averaged -0.195, -0.095, and -0.007. The factor version of the same model averaged -0.406, -0.203, and -0.003. Theorem 159 shows that the full contrasts $-2/5$ and $-1/5$, entered as treatment coefficients on the 0/1 recoding of Table 2.2, recover the four printed means.

4.4 Which scale the averages use

The question was which identification puts the fitted contrasts on the scale of Table 4.1. Three identifications were fit on the same draws.

The script, output, command, and commit are the same as the previous subsection. The marker multigroup model fixes the first loading at 1, equates loadings and intercepts, and frees four means. Two hundred replicates averaged -0.286, -0.134, and 0.007. One sample with 20,000 per group gave -0.277, -0.135, and 0.000. The marker population is -0.28, -0.14, and 0.

The fixed-measurement model fixes every loading at 0.7 and every intercept at 1. The same 20,000-per-group sample gave -0.398, -0.194, and 0.001. Two hundred replicates at $N = 240$ averaged -0.390, -0.206, and -0.010.

The variance-fixed model frees the loadings, fixes the factor variance at 0.50, and equates

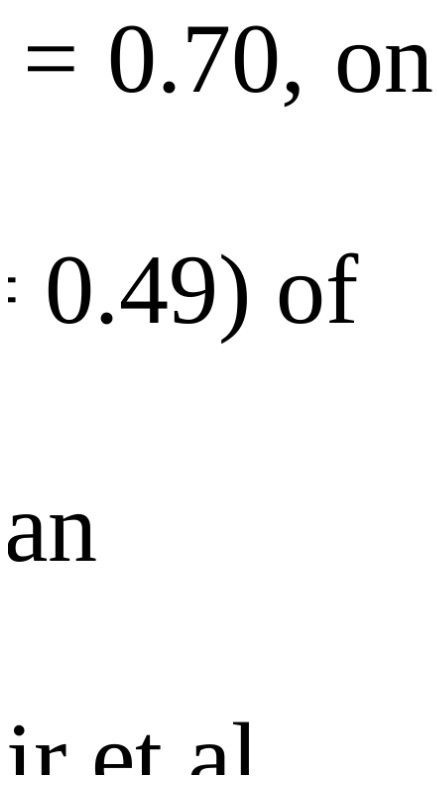


Figure 3: Population loadings 0.70, means $(1, 0.6, 0.8, 0.4)$, and factor variance 0.50.

loadings and intercepts. The large sample gave −0.400, −0.195, and 0.001.

Chapter III describes the marker constraints. Tables 4.1 and 4.2 print the generating scale. Both statements are on the page. The replay shows which constraint produces which scale. It does not decide which constraint Jeffries fit. The analysis syntax is not in Appendix A. That request is recorded as a help issue, linked in Section 6.

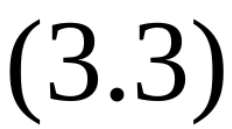
4.5 The first intercept token

The question was whether the text extract had dropped a digit from `i1 ~ 1*`. The ink is `1*`. The next three intercepts are `1*1`.

The artifact is `figures/appendix_intercept.png`, from the same script and commit. We did not find a second digit on that line. lavaan 0.6.17 does not parse `i1 ~ 1*`. simsem 0.5.16 was not run, so this does not say what simsem does with the token. Data generation in the replay used intercept 1 for every indicator.

5 What remains unresolved

The analysis syntax is unresolved. The marker sentence and the printed averages cannot be true of the same fitted contrast, on the population this appendix generates. Three candidates still fit the tables: loadings fixed at the data-generating value, factor variance fixed at 0.50, or a report of the contrasts after they have been divided by the estimated loading. The third was not fit





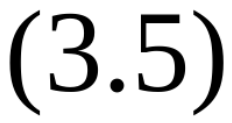


Figure 4: Equations (3.3)–(3.5). Each contrast is the mean of the groups coded +1, minus the mean of the groups coded −1.

here. Percent significant does not separate them. The missing script is issue https://github.com/alectionary/vibe-math/issues/33.

Convergence is unresolved. Table 4.5, as read from the text layer, reports 8,929 of 10,000 converged multigroup significance models at $N = 240$ and shares 45/45/5/5. All 200 of our marker fits in that cell converged, with `iter.max` of 1,000. We did not draw 15,000 simsem datasets, and we did not use the iteration cap of 10,000. The rate 89.29% is therefore not a number this replay tested.

The 200-replicate gaps in the third contrast's percent, in the unbalanced null cell and in the unbalanced true-difference cell, may shrink at 10,000 replicates. They may also be a feature of the marker standard errors. The replay cannot tell those apart at this size.

Jeffries and Bovaird (2026) is the journal version of this design. The PDF was not obtained. Routes are in Section 6. The dissertation comparison does not use that article.

The token `NA*i1 + 0.7*i1` was not passed through simsem. Our generator used the loading 0.7 once. If simsem treats the duplicated loading as something other than 0.7, the population would change. Nothing in the printed loading sentence suggests that.

Table 4.1
Multigroup Confirmatory Factor Analysis Results for True Difference Models

		Contrast 1			Contrast 2			Contrast 3		
Overall *N*	Condition	Avg. Est.	SE	% Sig.	Avg. Est.	SE	% Sig.	Avg. Est.	SE	% Sig.
240	1	-0.4046	0.0013	87.23	-0.2027	0.0013	35.57	-0.0010	0.0013	5.29
	2	-0.4045	0.0014	81.27	-0.2069	0.0014	31.63	0.0023	0.0014	5.66
	3	-0.4038	0.0021	49.23	-0.2181	0.0021	21.08	0.0054	0.0021	7.77
480	1	-0.4025	0.0009	99.23	-0.2015	0.0009	62.23	-0.0010	0.0009	5.15

Figure 5: Table 4.1, multigroup model, true difference. At $N = 240$ and equal groups the averages are -0.4046, -0.2027, and -0.0010.

Table 4.3
Multigroup Confirmatory Factor Analysis Results for Null Difference Models

		Contrast 1			Contrast 2			Contrast 3		
Overall *N*	Condition	Avg. Est.	SE	% Sig.	Avg. Est.	SE	% Sig.	Avg. Est.	SE	% Sig.
240	1	0.0011	0.0013	5.10	-0.0004	0.0013	4.77	-0.0021	0.0013	5.39
	2	0.0014	0.0014	5.02	-0.0044	0.0014	5.15	0.0017	0.0014	5.41
	3	-0.0028	0.0022	8.13	-0.0212	0.0023	8.59	-0.0012	0.0022	7.73
480	1	-0.0004	0.0009	5.22	-0.0006	0.0009	5.24	-0.0007	0.0009	5.14
	2	-0.0007	0.0010	4.86	-0.0001	0.0010	5.00	0.0014	0.0010	4.84
	3	-0.0002	0.0010	4.82	-0.0008	0.0010	4.91	0.0010	0.0010	4.63
960	1	-0.0011	0.0006	5.42	-0.0010	0.0006	5.15	-0.0007	0.0006	5.19
	2	-0.0008	0.0007	4.69	-0.0008	0.0007	4.84	0.0006	0.0007	4.89

Figure 6: Table 4.3, multigroup model, null difference. At $N = 240$ and shares 45/45/5/5 the percents significant are 8.13, 8.59, and 7.73.

6 Code and source availability

This section copies the ledger dated 2026-10-27. A row is a place we opened, not a recollection.

```
  "numeric"),
                                                    col.names = c("i1", "i2", "i3", "i4",
  "subgroup")))
  for (i in 1:length(cond1_n240_data_list)) {
    temp_data <- cond1_n240_data_list[[i]] |>
mutate(c1 = factor(case_when(subgroup == 1 ~ -1,
                                       subgroup == 3 ~
                                       -1, subgroup == 2
                                       ~ 1, subgroup == 4
                                     levels=c(-1, 1),
                                 labels = c("g1", "g2")),
           c2 = factor(case_when(subgroup == 1 ~ -1,
                                       subgroup == 2 ~
                                       -1, subgroup == 3
                                       ~ 1, subgroup == 4
                                     levels=c(-1, 1),
                                 labels = c("g1", "g2")),
           c3 = factor(case_when(subgroup == 1 ~ -1,
                                 subgroup == 4 ~
                                 -1, subgroup == 2
                                 ~ 1, subgroup == 3
                                 ~ 1),    levels = c(-1, 1),
                                 labels = c("g1", "g2")))
```

Figure 7: Appendix A. Groups 1 and 4 are coded −1 on `c3`, and groups 2 and 3 are coded +1. Codes `c1` and `c2` match Table 2.2.

Place	Date	Result
Appendix A of the dissertation PDF	2026-10-27	Generation syntax only. sha256 of the PDF is `a21df98c...fd469b3`
GitHub repository search for the dissertation title	2026-10-27	0 repositories
GitHub code search for Jeffries and orthogonal latent	2026-10-27	HTTP 429, so not an absence
OSF nodes with the dissertation title	2026-10-27	0 nodes
OSF `/v2/search/`	2026-10-27	HTTP 404
Sage page for DOI `10.1177/01466216261452255`	2026-10-27	HTTP 403
Unpaywall for that DOI	2026-10-27	PubMed abstract only, no PDF
Europe PMC for that DOI	2026-10-27	Not open access, no PDF
Wayback for that DOI	2026-10-27	No snapshot
PMC13179297	2026-10-27	HTTP 403, embargo named through 2027-05-15
lavaan 0.6.17, as named in the dissertation	2026-10-27	Installed and used
simsem 0.5.16 and dynamic 1.1.0	2026-10-27	Not installed. The authors' scripts are not those packages

The dissertation file itself was retrieved from the Wayback `id_` object dated 2025-05-11, snapshot `20250511025154`, of the DigitalCommons `viewcontent` URL for `dissunl` article 1215. The archive landing page is https://digitalcommons.unl.edu/dissunl/216/. Direct `viewcontent` returned HTTP 403. The help request for the analysis script is https://github.com/alectionary/vibe-math/issues/33. The fetch request for the 2026 article is https://github.com/alectionary/vibe-math/issues/34.

```
# ---- Models of N = 240 ----
pacman::p_load(here, tidyverse, simsem, lavaan, purrr)
set.seed(1313)
# ---- Significance Model ----
## ---- Model Syntax ----
# 4 item model
pop_model_cond1 <- "
eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4
eta ~~ .5*eta
i1 ~~ 0.51*i1
i1 ~ 1*
i2 ~~ 0.51*i2
i2 ~ 1*1
i3 ~~ 0.51*i3
```

Figure 8: Appendix A population model. The first intercept is written `1*`. The loading line contains both `NA*i1` and `0.7*i1`.

7 Next steps

The next check is the analysis function that produced Tables 4.1 and 4.2. With that syntax, refit one large sample and see whether the contrasts land near -0.40 or near -0.28. The same script would show the iteration settings behind the 8,929/10,000 convergence count. The 2026 article would show whether the published paper uses the same tables. Neither file is required to trust the orthogonality proof, which uses only the printed codes and means.

8 What we proved

The codes, in group order $(1, 2, 3, 4)$, are

$$c_1 = (-1, 1, -1, 1), \qquad c_2 = (-1, -1, 1, 1), \qquad c_3 = (1, -1, -1, 1).$$

The weighted product is $\sum_{g=1}^{4} n_g c_{jg} c_{kg}$. For equal sizes the three products are zero (107). For sizes (a, a, b, b) the products of c_1 with c_2 and of c_2 with c_3 stay zero, and the product of c_1 with c_3 equals $2(b - a)$ (68). The column sums are zero (99), which is the contrast property that the codes do not confound the grand mean.

A defined contrast is the average of the means coded $+1$, minus the average of the means coded -1. On $\kappa = (1, 3/5, 4/5, 2/5)$ the three contrasts are $-2/5$, $-1/5$, and 0 (123). The third is zero while $\kappa_1 \neq \kappa_2$ (127). Multiplying every mean by the loading $7/10$ multiplies the contrasts by $7/10$, giving $-7/25$, $-7/50$, and 0 (131). The treatment-contrast form with coefficients $-2/5$, $-1/5$, and 0 reproduces κ (159).

These identities are specific to the printed codes and the printed means. They are not a theorem about every four-group contrast. The proofs are rational arithmetic. The `ring` proofs depend on `propext`, `Classical.choice`, and `Quot.sound`. The `native_decide` proofs also depend on `Lean.ofReduceBool`.

9 Artifacts

- Products and population contrasts. File `LatentContrasts/Orthogonality.lean`. Command: `lakebuild` in `formalization/latent-contrasts`.
- Main grid of marker and MIMIC fits. File `simulation/out/monte_carlo.csv`. Command: `Rscriptmonte_carlo.R200100out`.
- Identification toggle and the large sample. File `simulation/out/identification_check.csv`. Command: `Rscriptidentification_check.R200out`.
- Page crops. Files `paper/figures/*.png`. Command: `python3paper/figures/crop_pages.py`.

The dissertation PDF is local and gitignored. Its sha256 is `a21df98c20401491e6b4a1d00a21b6f7cee5551a00c04a89fb414ec6cfd469b3`. The crop command expects that file at `local-extraction/literature/unl-sem/retrieval-repair/UNL-SEM-D055.pdf`.